PATH Report

Auto-generated summary of PATH outputs

PATH is a decision-support tool to help identify where to prioritize hillslope-level treatment given the user selected catchment, treatments, treatment costs, and sediment thresholds. This report provides a summary of PATH's outputs including tables, plots, and maps of the selected hillslopes and treatments. Section 3 provides additional analysis of the sediment thresholds and their impact on treatment selection and costs.

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1 Parameters

The following parameters were used for this PATH run:

Parameter	Value
Hillslope Sediment Yield Threshold (tons/acre)	15.0
Outlet Sediment Discharge Threshold (tons)	200.0
Treatments	['0.5 tons/acre', '1 tons/acre', '2 tons/acre']
Treatment Cost ($/acre)	[2475.0, 2475.0, 2475.0]

2 Run Results

2.1 Static Map of Selected Hillslopes

The map below shows the selected hillslopes by treatment. Hillslopes outlined in red exceeded the hillslope sediment yield (Sdyd) threshold even after treatment.

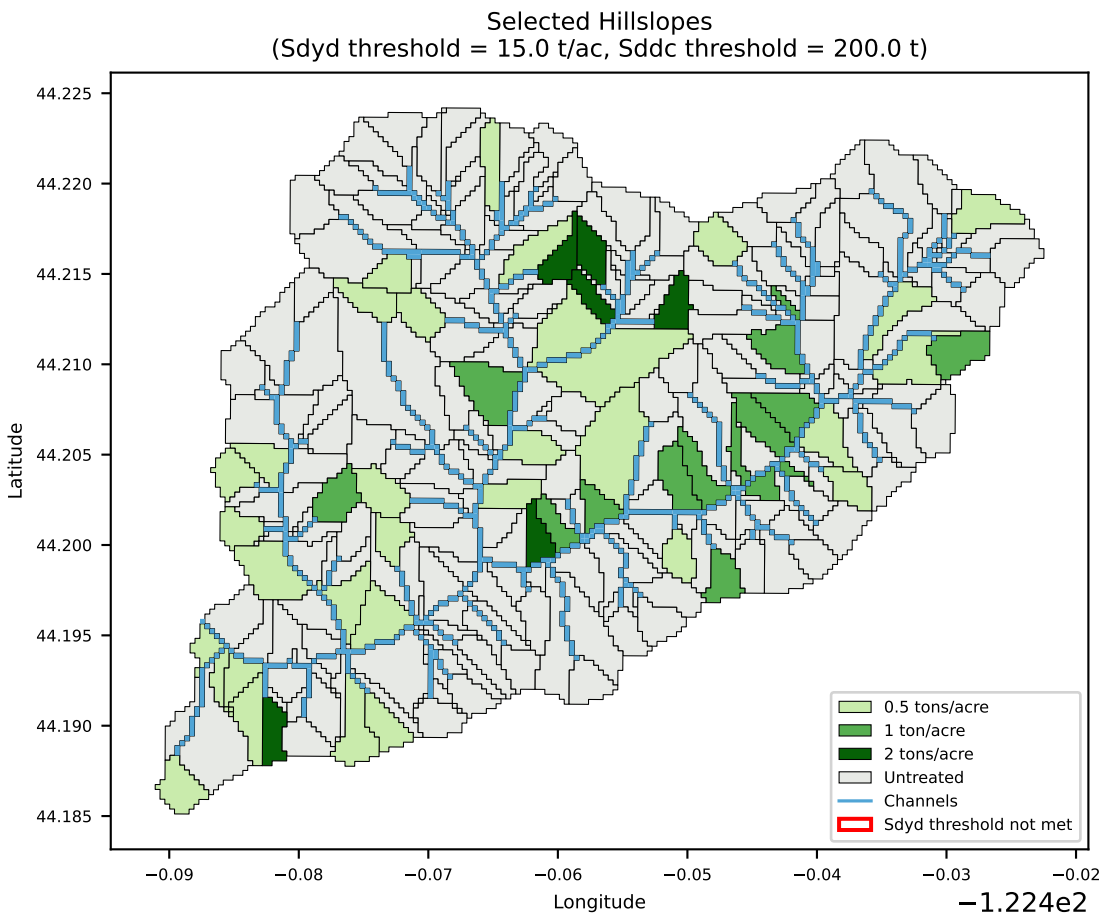


Figure 1: Selected hillslopes by treatment assignment. Red outlines indicate hillslopes where the sediment yield threshold could not be met.

2.2 Key Outputs

Metric	Value
Total cost ($)	$634,280.01
Total outlet sediment discharge reduction (tons)	164.771
Final outlet sediment discharge (tons)	199.974

2.3 Combined Hillslope Treatment and Final Sdyd

The table below combines treatment assignment and final hillslope sediment yield (Sdyd). Any hillslope present in the final Sdyd output but missing from treatment assignments is labeled as `No treatment`. Hillslopes with exceeded Sdyd threshold are flagged.

	wepp_id	final_Sdyd	treatment	sdyd_threshold_exceeded
0	1	7.676414	0.5 tons/acre	False
1	2	3.221717	No treatment	False
2	3	3.116150	0.5 tons/acre	False
3	5	2.953308	0.5 tons/acre	False
4	7	4.044105	No treatment	False
5	10	8.408968	2 tons/acre	False
6	11	9.767267	0.5 tons/acre	False
7	18	2.487260	0.5 tons/acre	False
8	23	9.466524	0.5 tons/acre	False
9	27	4.218379	No treatment	False
10	28	2.762775	0.5 tons/acre	False
11	29	7.349570	0.5 tons/acre	False
12	30	3.863297	0.5 tons/acre	False
13	34	4.792485	1 tons/acre	False
14	39	6.324834	0.5 tons/acre	False
15	44	6.985992	0.5 tons/acre	False
16	54	5.617949	0.5 tons/acre	False
17	57	3.793868	No treatment	False
18	58	3.947096	0.5 tons/acre	False
19	65	4.281842	No treatment	False
20	69	3.898494	No treatment	False
21	77	5.338186	No treatment	False
22	83	6.222495	0.5 tons/acre	False
23	86	2.567280	0.5 tons/acre	False
24	89	3.119259	2 tons/acre	False
25	97	0.449298	1 tons/acre	False
26	104	5.135867	0.5 tons/acre	False
27	106	2.816693	No treatment	False
28	110	2.463552	0.5 tons/acre	False
29	116	0.414211	1 tons/acre	False
30	120	4.358051	1 tons/acre	False
31	126	6.155920	0.5 tons/acre	False
32	131	3.985556	No treatment	False
33	137	7.731268	0.5 tons/acre	False
34	138	1.764109	0.5 tons/acre	False
35	140	4.627937	1 tons/acre	False
36	141	8.952783	0.5 tons/acre	False
37	143	2.452836	0.5 tons/acre	False
38	149	4.066561	2 tons/acre	False
39	150	2.070321	2 tons/acre	False
40	154	3.065280	1 tons/acre	False
41	155	3.222005	1 tons/acre	False
42	162	5.124626	2 tons/acre	False

	wepp_id	final_Sdyd	treatment	sdyd_threshold_exceeded
43	174	3.937209	2 tons/acre	False
44	176	2.262656	0.5 tons/acre	False
45	180	1.502525	1 tons/acre	False
46	187	2.647163	No treatment	False
47	192	5.715146	1 tons/acre	False
48	209	1.796213	0.5 tons/acre	False
49	210	1.295640	1 tons/acre	False
50	211	3.084481	0.5 tons/acre	False
51	214	4.992821	0.5 tons/acre	False
52	223	6.281164	0.5 tons/acre	False
53	226	4.726267	No treatment	False
54	250	1.073202	1 tons/acre	False
55	258	1.632564	0.5 tons/acre	False
56	263	0.157153	0.5 tons/acre	False
57	264	2.429335	No treatment	False
58	265	4.342839	1 tons/acre	False
59	266	0.903674	0.5 tons/acre	False
60	268	3.739179	No treatment	False
61	295	7.288826	0.5 tons/acre	False
62	299	4.456133	No treatment	False

	treatment	n_sites	cost_per_acre ($)
0	0.5 tons/acre	31	2475.0
1	1 tons/acre	12	2475.0
2	2 tons/acre	6	2475.0
3	No treatment	14	NaN

3 Sediment Threshold Analysis

This section provides additional analysis of the Hillslope Sediment Yield and Outlet Sediment Discharge thresholds and their impact on treatment selection and costs.

3.1 Hillslope Selection Maps at Threshold Percentiles

The following maps show the hillslopes selected for treatment when the combined sediment thresholds (Sdyd and Sddc) are set at the 25th, 50th, and 75th percentiles of total treatment cost across all evaluated threshold combinations.

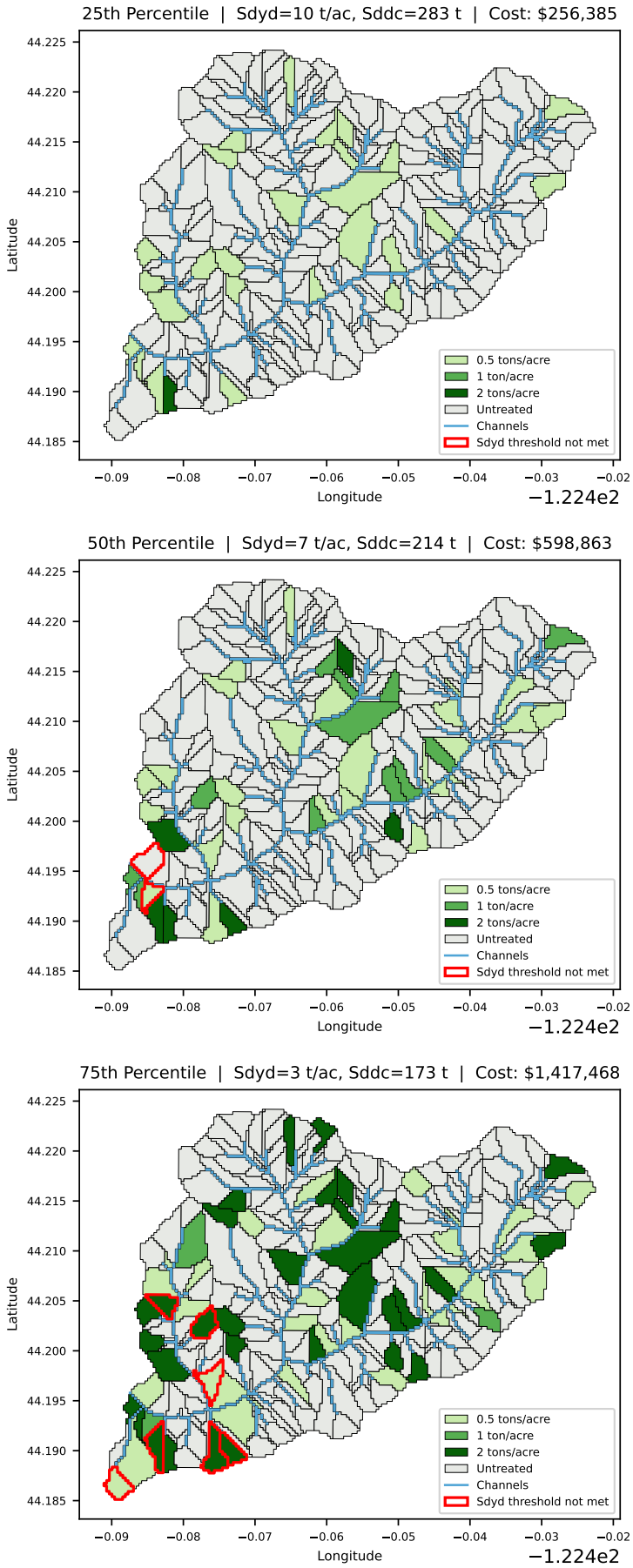
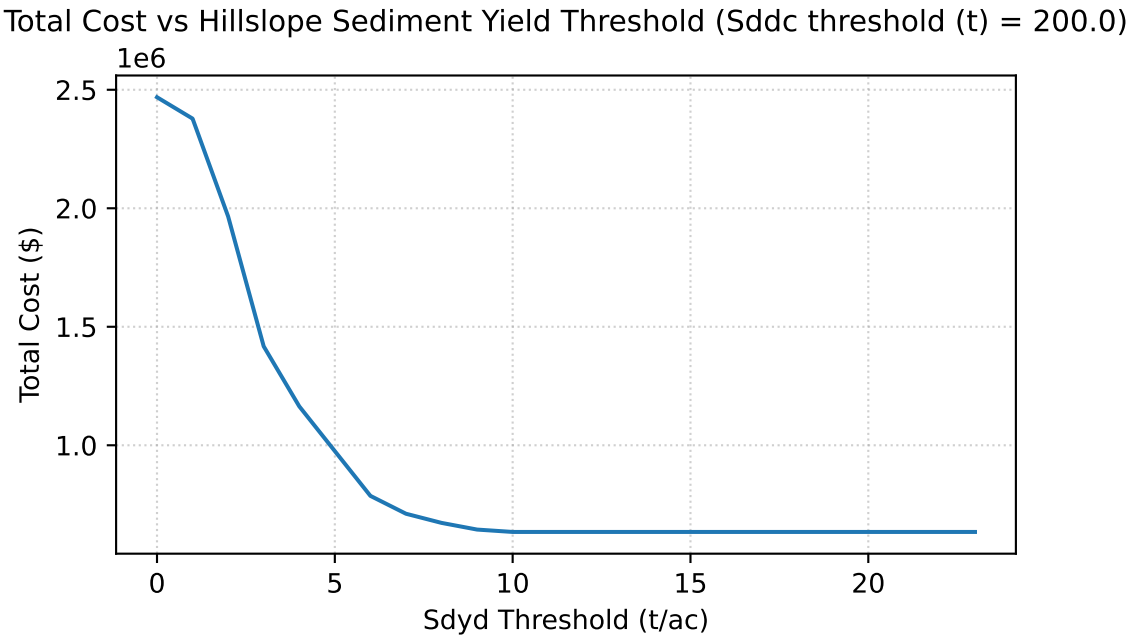


Figure 2: Selected hillslopes at the 25th, 50th, and 75th cost percentiles of the threshold sweep.

3.2 Cost vs. Threshold Visualizations

The following plots show the relationship between total cost and the two sediment thresholds.



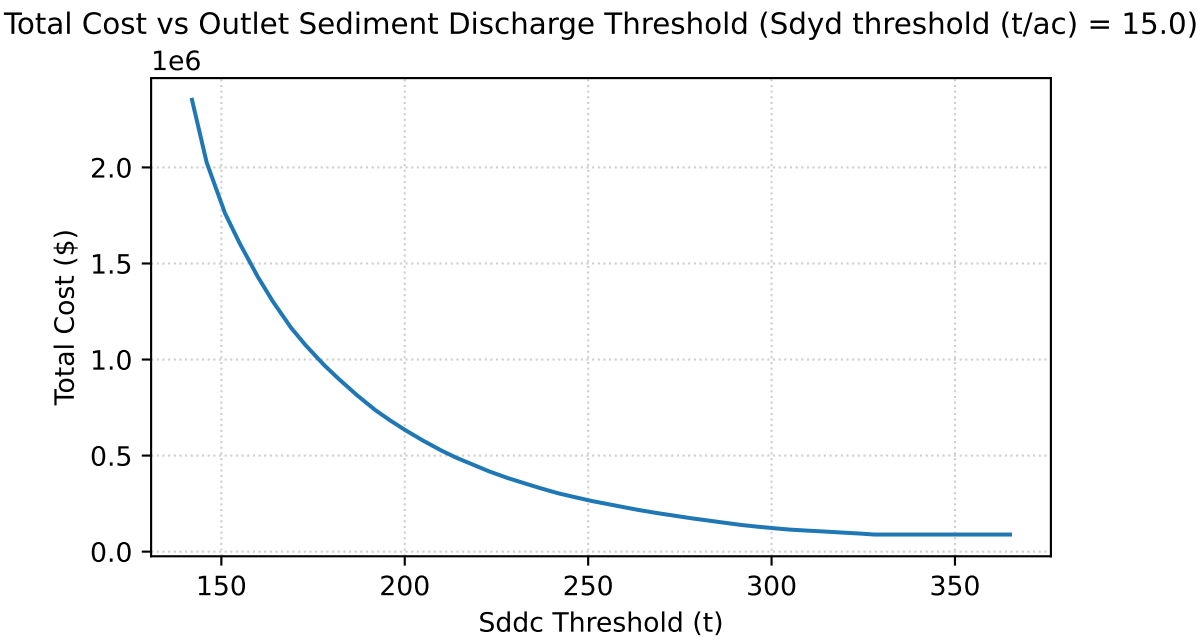


Figure 3: Total treatment cost as a function of each sediment threshold.

3.3 Total Cost Surface

The surface below shows total treatment cost across all evaluated combinations of the two sediment thresholds. The red marker (+) indicates the user-selected thresholds. The blue (25th), green (50th), and orange (75th) markers indicate the three cost percentile scenarios shown in the maps above.

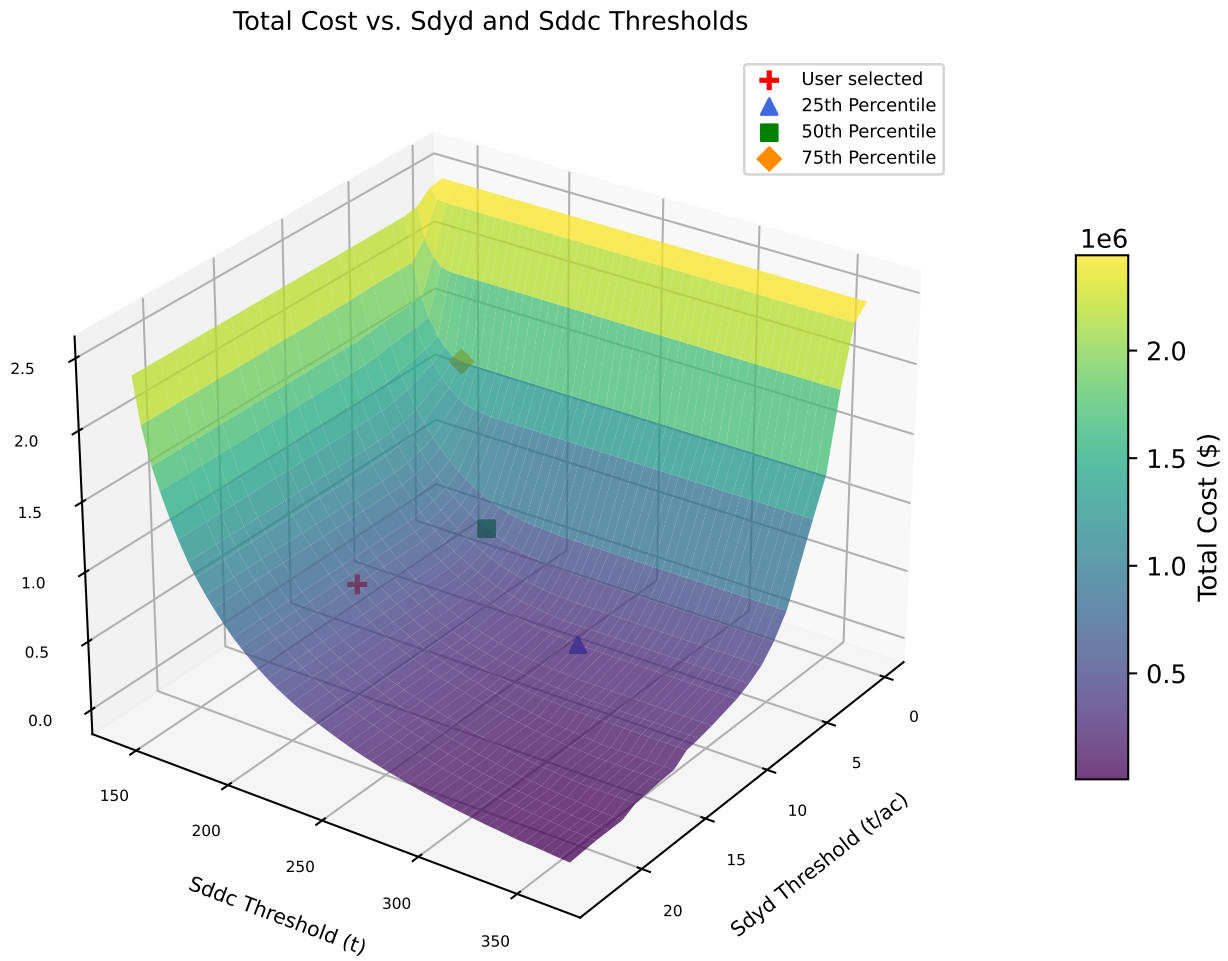


Figure 4: 3D cost surface with user-selected and percentile threshold markers.